

Linearization

Spatial discretization

LCHS or Schrödingerization

Hamiltonian embedding

Nonlinear PDE

Linear Hyperbolic PDE

Linear ODE

Sparse Hamiltonian simulation

Problem Hamiltonian H

Encoding (e.g., std. binary, one-hot, unary, etc.)

&

Decomposition (e.g., Pauli basis, Bell basis, etc.)

Embedding Hamiltonian $\tilde{H}$

Hardware-efficient simulation